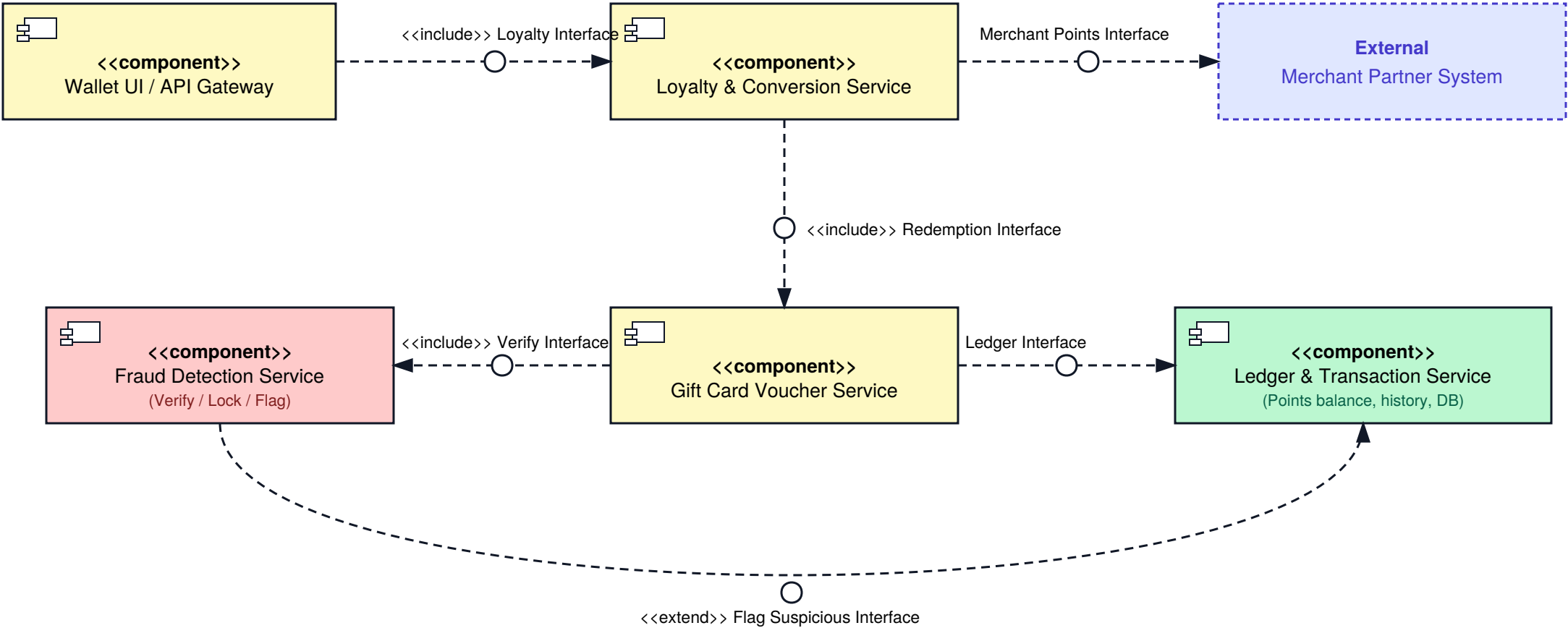


# Digital Gift Card & Loyalty Point Exchange — Microservices Component Diagram

Problem Statement #34 — PES1UG24CS571



## Actors (from Lab 1 use-case diagram)

- Account Holder — interacts via Wallet UI / API Gateway
- System Admin — monitors ledger & fraud alerts (not shown as separate component; accesses Ledger & Fraud services)
- Merchant Partner — external system supplying loyalty points, shown as external component

## Traceability to Lab 1 Use Cases

Convert Loyalty Points → Gift Card = Loyalty & Conversion Service + Gift Card Voucher Service (UC uses «include» Generate Digital Gift Card)

Redeem Gift Card at Merchant (UC-04) = Voucher Service «include» Verify & Lock Voucher (Fraud Detection) «include» Update Ledger & Points Balance (Ledger Service)